

LEEDS BRADFORD, United Kingdom

WMO: 033463

Lat: 53.866N

Lon: 1.661W

Elev: 681

StdP: 14.34

Time Zone: 0.00 (EUW)

Period: 95-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) 26.2	(c) 28.2	(d) 22.6	(e) 17.4	(f) 31.1	(g) 24.5	(h) 19.1	(i) 31.8	(j) 38.8	(k) 47.1	(l) 33.2	(m) 44.7	(n) 8.2	(o) 280	(p) 0.665

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 13.4	(c) 75.2	(d) 64.0	(e) 71.6	(f) 62.0	(g) 68.2	(h) 60.4	(i) 65.2	(j) 72.1	(k) 63.3	(l) 69.4	(m) 61.6	(n) 67.1	(o) 9.3	(p) 180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
62.6	87.2	68.4	60.9	81.8	65.9	59.2	77.0	63.8	30.5	71.9	29.1	69.6	27.9	66.7	77.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a) 28.4	(b) 24.7	(c) 21.6	(e) 22.4	(f) 80.2	(g) 3.0	(h) 3.2	(i) 20.2	(j) 82.5	(k) 18.5	(l) 84.3	(m) 16.9	(n) 86.1	(o) 14.7	(p) 88.4		
			DB													
			WB													
				22.2	70.1	3.1	3.4	20.0	72.6	18.2	74.6	16.5	76.5	14.2	78.9	

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	48.2	38.7	39.3	41.8	45.6	50.9	55.8	59.7	59.2	55.0	49.5	42.9	39.5
	DBStd	9.07	5.19	5.41	5.25	5.13	5.24	4.62	4.49	4.19	4.09	4.91	5.15	5.83
	HDD50	1734	349	302	259	148	52	5	0	0	5	69	217	328
	HDD65	6152	814	721	721	581	438	277	176	188	300	482	662	792
	CDD50	1080	0	1	3	17	79	180	300	286	156	52	5	1
	CDD65	23	0	0	0	0	0	3	11	8	1	0	0	0
	CDH74	145	0	0	0	0	2	22	73	44	3	1	0	0
	CDH80	11	0	0	0	0	0	1	6	4	0	0	0	0
(14) Wind	WSAvg	10.5	12.1	12.2	11.5	10.2	9.9	9.3	9.3	9.2	9.4	10.3	11.0	11.5
(15) Precipitation	PrecAvg	34.5	3.5	2.7	2.8	2.5	2.3	2.6	2.2	2.7	2.9	3.0	3.5	3.6
	PrecMax	43.8	7.1	6.6	5.5	5.2	4.9	7.7	5.5	6.3	8.3	6.7	6.7	7.7
	PrecMin	25.4	0.2	0.3	0.5	0.3	0.3	0.4	0.3	0.3	0.3	0.5	1.3	1.2
	PrecStd	4.8	1.6	1.5	1.3	1.3	1.2	1.7	1.2	1.3	1.9	1.6	1.7	1.7
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.4	55.5	57.6	66.2	73.1	77.4	80.3	79.1	73.2	64.7	57.4	54.0
		MCWB	50.7	50.4	50.1	55.2	61.9	65.3	65.2	64.0	64.1	59.4	54.8	51.5
	2%	DB	50.3	51.8	54.0	60.6	68.0	71.6	75.5	73.3	67.8	60.9	55.1	52.0
		MCWB	48.2	48.6	48.7	52.4	58.4	61.6	64.3	63.3	60.2	57.1	52.7	50.0
	5%	DB	48.4	49.6	51.8	57.1	63.0	67.9	71.8	69.5	64.3	58.8	52.2	50.1
		MCWB	47.0	47.2	47.5	50.3	55.2	59.4	62.5	61.8	58.6	55.8	50.0	48.5
	10%	DB	46.5	46.8	49.9	53.9	59.4	64.5	68.2	66.3	62.2	57.0	50.3	48.2
		MCWB	45.1	44.9	46.5	48.3	53.1	57.3	60.7	59.9	56.9	54.2	48.8	47.0
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	51.2	51.8	52.3	57.1	62.7	67.1	67.9	66.8	64.8	60.7	55.7	52.8
		MCDB	52.8	54.0	56.2	61.8	71.1	74.5	76.3	73.7	70.2	64.2	57.5	54.4
	2%	WB	48.9	49.3	50.1	53.5	58.9	62.8	65.4	64.4	61.7	58.0	53.3	50.9
		MCDB	49.9	51.0	52.9	58.9	65.7	69.1	72.8	70.2	66.2	60.3	54.9	52.1
	5%	WB	47.1	47.2	48.3	51.2	56.5	60.2	63.3	62.7	59.7	56.0	51.2	48.7
		MCDB	48.1	48.9	50.8	56.0	62.2	65.9	69.9	68.6	63.3	58.2	52.6	49.7
	10%	WB	45.1	45.2	46.6	49.3	54.1	58.3	61.5	61.1	57.9	54.3	49.2	46.6
		MCDB	46.1	46.8	49.3	53.3	59.1	63.3	67.3	65.7	61.0	56.4	50.3	47.9
(35) Mean Daily Temperature Range	5% DB	MDBR	7.4	8.6	10.7	12.8	13.4	13.3	13.4	12.5	11.5	9.7	8.2	7.7
		MCDBR	9.7	10.7	13.3	16.9	19.0	19.3	19.4	16.9	15.0	11.3	9.6	9.7
		MCWBR	9.6	8.9	9.1	10.4	11.5	11.4	10.3	8.9	9.1	8.5	9.1	9.2
	5% WB	MCDBR	9.5	9.8	12.0	15.5	17.8	17.1	17.4	14.9	13.3	10.8	9.6	9.6
		MCWBR	9.7	9.0	9.3	10.7	11.5	11.4	10.5	9.0	9.5	9.0	9.5	9.5
(40) Clear-Sky Solar Irradiance	taub		0.314	0.329	0.358	0.384	0.384	0.387	0.387	0.374	0.356	0.339	0.314	0.303
	taud		2.331	2.388	2.314	2.260	2.296	2.313	2.314	2.367	2.426	2.448	2.397	2.311
	Ebn at Noon		204	243	259	267	273	273	271	268	259	237	204	182
	Edh at Noon		18	24	31	38	38	38	38	34	28	22	17	16
(44) All-Sky Solar Radiation	RadAvg		201	411	757	1182	1500	1541	1491	1227	893	502	248	157
	RadStd		22	47	89	133	96	141	146	80	79	44	26	16

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (66 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page